



IMAGE NOISE MODELS AND NOISE REMOVAL

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INTRODUCTION

TO IMAGE NOISE

Digital images are often degraded by noise during acquisition and transmission. This unwanted variation in pixel intensity reduces visual quality and accuracy, making noise removal a critical preprocessing step.



→ What is Image Noise?

Random fluctuations in brightness or color values that were not present in the original scene. Noise manifests as grainy texture or speckles across the image.



● The Problem

- Noise causes random disturbances in images
- It corrupts pixel data and reduces quality
- Common sources:
 - Sensor errors
 - Poor lighting conditions
 - Transmission interference

● Why It Matters

- Noise affects image analysis and processing
- Reduces accuracy of results
- Noise removal helps to:
 - Improve computer vision performance
 - Enhance pattern recognition
 - Support medical/diagnostic imaging

TYPES OF NOISE

Image noise can be classified into different types based on distribution and behavior

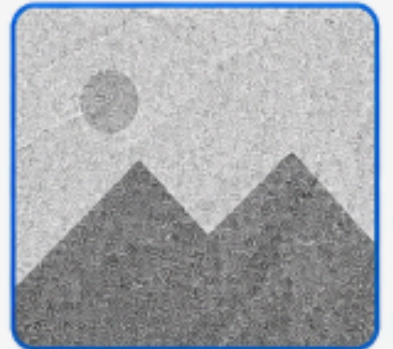
Main Types:

- Gaussian Noise → smooth, grain-like variation
- Salt & Pepper Noise → black & white pixel spikes
- Uniform Noise → evenly distributed random values



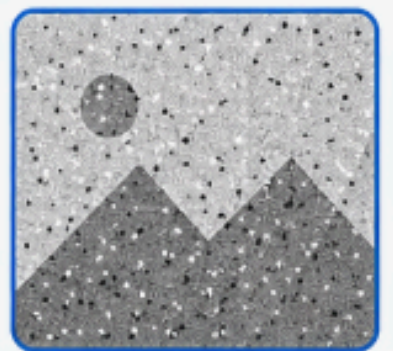
GAUSSIAN NOISE

Smooth, grain-like variations that affect all pixels.



SALT & PEPPER NOISE

Black and white pixel spikes appearing randomly.



UNIFORM NOISE

Random pixel values evenly distributed within a fixed range.



GAUSSIAN NOISE

Distribution Pattern

- Follows a normal (bell curve) distribution
- Most values are near the mean (μ)

Cause

- Caused by electronic disturbances in sensors and circuits

Characteristics

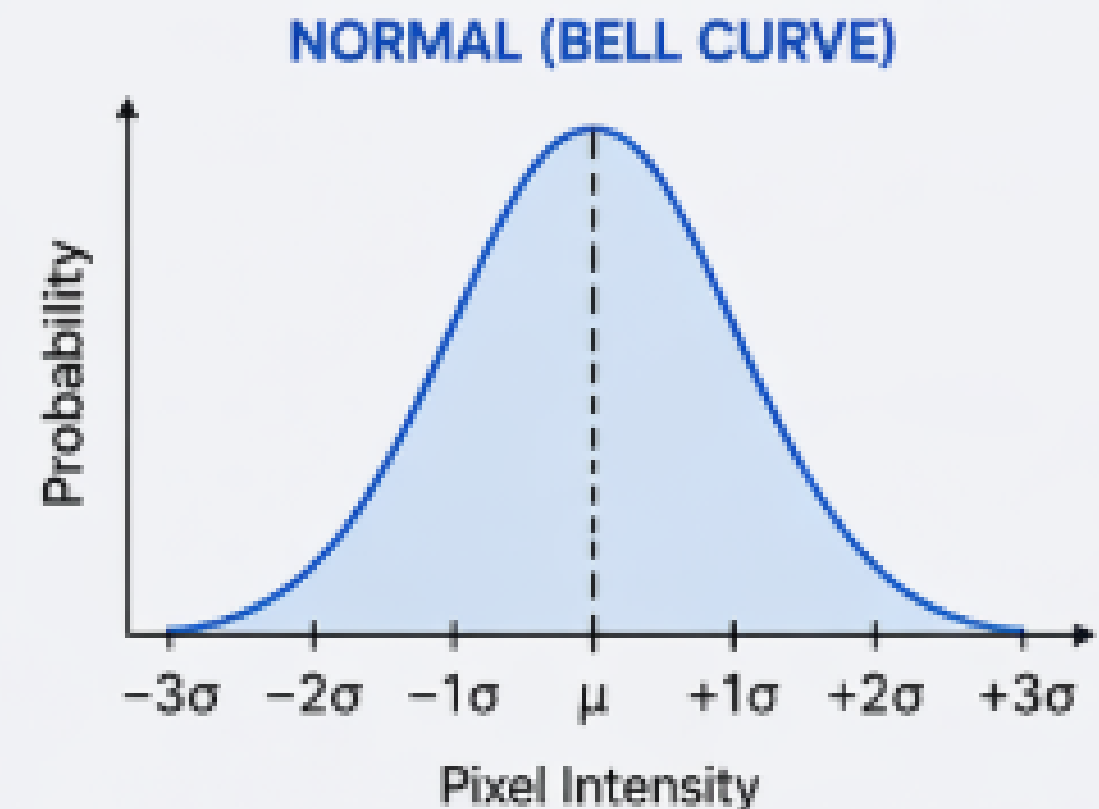
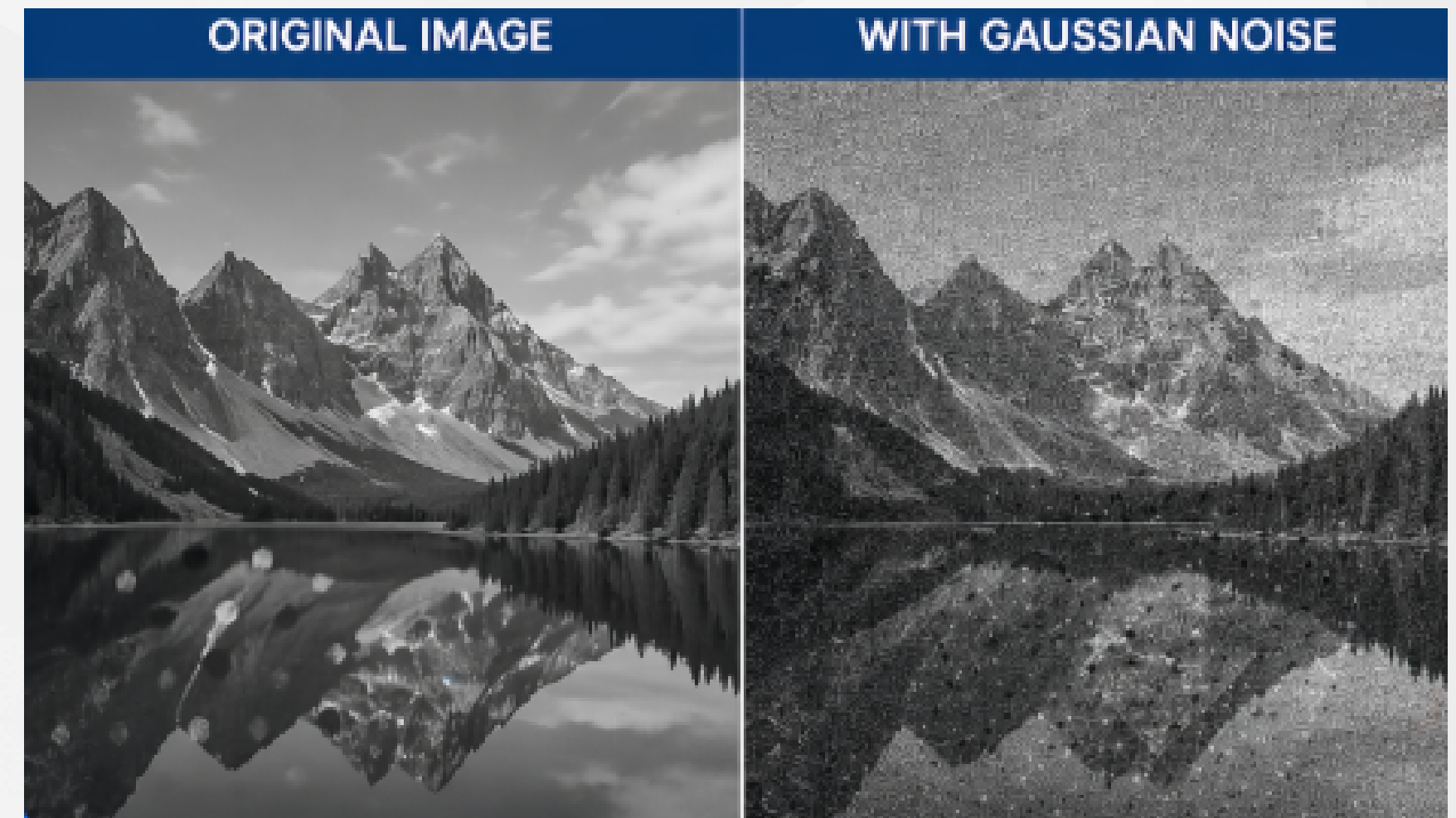
- Affects all pixels with varying intensity
- Spread across the entire image uniformly

Visual Appearance

- Appears as smooth, grain-like texture

Mathematical Model

- Defined by mean (μ) and standard deviation (σ)



SALT & PEPPER NOISE

Impulse Noise

- Also called impulse noise
- Sudden changes in pixel intensity

Appearance

- Appears as random black & white pixels
- Creates a distinct speckled pattern

Cause

- Caused by transmission errors
- Faulty memory or conversion issues

Characteristics

- Affects only some pixels
- Produces high contrast (0 or 255 values)

Original Image



Image with Salt and Pepper Noise

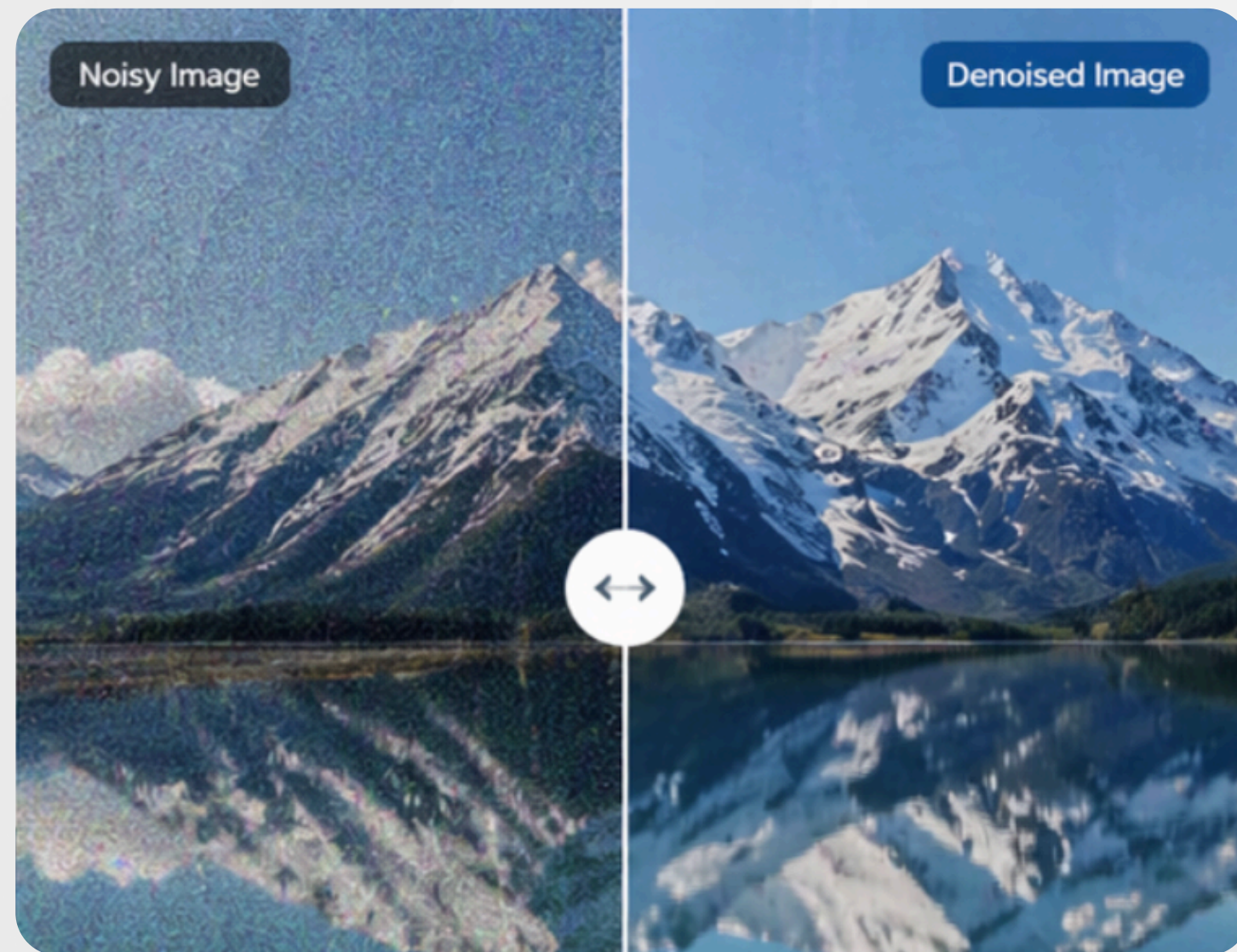


Salt and Pepper Noise



NOISE REMOVAL TECHNIQUES

Noise is reduced using filtering techniques. Common filters include mean and median filters. The goal is to improve image quality for better analysis and processing accuracy.



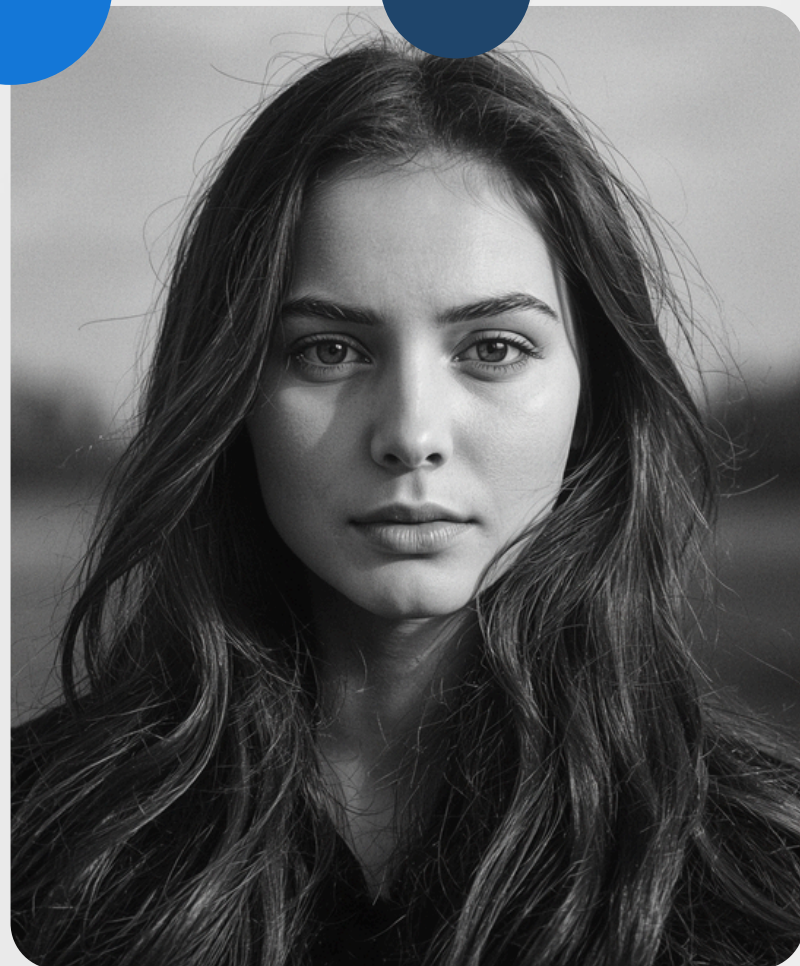
Mean Filter

Replaces each pixel with the average of its neighbors. Smooths the image and reduces noise effectively, but has the limitation of blurring sharp edges.



Median Filter

Replaces each pixel with the median value of neighbors. Highly effective for impulse noise and preserves edges better than mean filtering.



MEAN FILTER

The mean filter is a simple spatial filtering technique used for noise reduction. It works by replacing each pixel value with the average of its neighboring pixels within a defined window. This averaging operation effectively smooths the image and reduces random noise variations.

→ How It Works

Replaces each pixel with the average value of surrounding neighbors. Effectively reduces Gaussian noise by smoothing intensity variations across the image.

→ Limitations

While effective at noise reduction, mean filtering causes edge blurring. Sharp transitions between regions become softened, reducing image detail and clarity.



MEDIAN FILTER

The median filter replaces each pixel with the median value of neighboring pixels. It is highly effective for removing impulse noise (salt and pepper noise) while preserving important edge details better than mean filtering approaches.

➔ Before Filtering

Image degraded with salt and pepper noise showing random black and white pixel artifacts that obscure underlying details.

➔ After Filtering

Same image after median filtering showing significant noise reduction with sharp edges and fine details preserved intact.

Original



Adaptive Filtered



ADAPTIVE FILTERS

Adaptive filters adjust their behavior based on local image region characteristics, providing superior noise removal compared to fixed filters.

● Key Characteristics

- Analyzes local statistics dynamically
- Varies filter window size as needed
- Preserves fine details and edges

● Adaptive Median Filter

- Expands window until valid median found
- Handles varying noise densities
- Outperforms standard median filter

CONCLUSION

Understanding noise models and applying appropriate filtering techniques are essential skills in digital image processing. The right approach ensures cleaner, more accurate results.

● Key Takeaways

Noise models help understand image degradation patterns. Various filtering techniques exist for different noise types. Median and adaptive filters preserve edges while removing noise effectively.

● Practical Applications

Filter choice depends on noise type and application requirements. Proper noise removal significantly improves image processing accuracy. Always analyze noise characteristics before selecting a filter.



THANK YOU!

Thank you for your attention!

